Energy Demands of 802.15.4/ZigBee Communication with IRIS Sensor Motes

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Energy Demands of 802.15.4/ZigBee Communication with IRIS Sensor Motes

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***Abstract*—Limited energy sources of nodes in wireless sensor networks require a careful consideration of energy consumption of all processes during the sensors deployment. To analyze energy consumption and to predict a lifetime of a network, a comprehensive energy model based on commercial products is necessary. Therefore, we have focused on the analysis of energy consumption during RF communication with particular 802.15.4 compliant nodes. We have designed experimental testbed and explored the scenario of node association and data transmission. For the measurement we used IRIS sensor nodes with 802.15.4/Zigbee protocol stack and shunt connection. The results show that energy consumption cannot be calculated only from datasheet values of current drain and length of a packet but intervals of listening and waiting between transmissions play important role as well.**

***Keywords*—Energy, 802.15.4/ZigBee, WSN, RF communication, measurements, IRIS sensor nodes.**

I. Introduction

WIRELESS sensor networks (WSNs) are intended to be an information technology for data gathering from environments with specific characteristics or requirements. In some applications, WSNs are just more economical option, in others the only possibility of meeting the determined demands. The applications can vary from home automation data gathering system to applications in very hostile and harsh environment such as active volcano or mines. In most cases, sensor nodes are battery operated and maintenance related to battery replacement means extra costs and additional problems especially in inaccessible or restricted areas [1][2].

Limitation in energy sources is one of the constrains of WSN (besides computational, memory and bandwidth resources). When energy is depleted nodes lose their sensing and communication functionalities, become inactive, dead. With a non-uniform energy depletion in the network, some of the nodes are crucial for the whole network or its certain part and if those nodes are dead the whole dependent subnetwork (cluster) is disconnected. Therefore, to improve network reliability and prolong network lifetime, certain techniques of communication, low-powered nodes and energy aware network topologies have to be involved in a network design.

Improvements and various energy saving techniques can be employed in the whole protocol stack. Such techniques include MAC sleeping schemas, network routing protocols, localization, security and other energy aware services and functions [3][4].

IEEE standard 802.15.4 was accepted as a standard for short-range, low-energy Wireless Personal Area Networks (WPANs). This standard describes the Physical and Medium Access Control Layer parameters such as frequency range, modulation, bit rate, time intervals, topology etc. [5]. Following this standard, several companies designed and produced 802.15.4 compliant devices such as IRIS from CrossBow, WaspMote from Libelium or Tiny Nodes from Berkeley.

In order to assure a full compatibility of different WSN products, specification of two bottom layers is not sufficient. Therefore, ZigBee alliance extended the 802.15.4 standard into a 802.15.4/ZigBee specification [6], which covers the whole protocol stack up to application layer framework. This specification allows products from different manufactures, following the specification, talk to each other.

Since then, there have been designed several specifications defining protocol stack for WSNs such as proprietary XMesh protocol from Crossbow [7]. However, Zigbee is still the most common standard in commercial products and thus we focus on it.

Our investigation is aimed to find real energy consumption of a WSN node during network operations. We measure current drain of a node circuits and base on it we try to explore phases and energy depletion during those phases related to communication between two nodes. To relate results to some real devices we decided to use IRIS sensor nodes from CrossBow with implemented ZigBee protocol stack. The obtained results allow us to see properly the consumption of node, time duration of individual phases and their current drain. Base on that we can perform more accurate energy analysis and design more appropriate energy model of communication. Moreover, we can optimize protocols and their time-schemes to save precious energy, which as a consequence results in longer WSN network lifetime. Therefore, we designed a measurement testbed composed of two communication nodes and current measurement setup in order to perform necessary measurements. The current measurement setup is due to small values in order of mA based on a shunt resistor and voltage amplifier to get appropriate results. For data acquisition and processing we used a sampling card and Matlab simulation tool.

The rest of the paper is organized as it follows: Section 2 discusses briefly energy constraints in WSN, Section 3 describes proposed experimental testbed and devices used. In the following section, Section 4, the results of measurements and comments are presented. The last Section 5 summarizes

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conclusions of our work and measurements.

II. Energy in WSN

Energy related questions are a permanent topic of research in WSN. Energy is a precious resource in WSN like bandwidth and computing power. This limitation is so important for entire network design that a new standard had to be introduced. It was not possible to describe low-power sensor nodes and their efficient communication by any other standard for wireless communication and thus, 802.15.4 standard was proposed. The crucial aspect of WSN is that nodes are generally equipped just with one battery unit and when the source is depleted the node is death. Certain types of nodes in a network can have extended functionality (as routing) and therefore they need more energy (perform more tasks and are up for a longer period). In ZigBee networks those nodes can be powered from mains supply. However, considering certain remote monitoring application without possibility of permanent power source for those nodes, the network has to handle this situation employing different approach. In certain applications, energy harvesting from the node surrounding is a good option. In this cases solar panels, wind mils or other energy harvesters (mechanical, acoustic, ultrasonic, thermoelectric) can be used [8] [9]. Interesting approach making use of electromagnetic energy of useless RF signals was introduced by L.Tang and Ch. Guy in [10].

If we have no possibility of mains supply or energy harvesting we have to guarantee that provided energy sources last till the end of the determined application lifetime. To assure that, minimal consumption is necessary. There are several approaches how to reduce energy during the node lifetime. Besides energy efficient design of all the hardware components of a WSN node, optimization of all processes in the network from the energy point of view is essential.

If we want to evaluate energy demands of a certain algorithm, process or if we want to estimate the lifetime of a network, we have to use an energy model of the node and network and perform appropriate simulations. The simulation results are as precise as the model is closed to the real situation. Therefore, several models have been proposed [11] [12].

There are three main consumers of energy regarding WSN node: computing power of microcontroller (MCU), RF communication and sensing. When we add up all those components we get an expression of entire energy consumption:

$$E = E_{\mu p} + E_{RF} + E_{sensor} \tag{1}$$

$E_{\mu p}$ is the energy consumed by microprocessor, E_{RF} energy drained by transceiver circuits (transmission and reception both together) and E_{sensor} stands for the energy needed to power all the sensors required by application task.

Energy of sensing is highly dependent on the type of sensor used. It can range from units of mA to hundreds of mA and it can be hardly changed or optimized from the network point of view. Reduction of sensing power is possible only by prolonging the interval between sensing phases.

III. Experimental testbed

In order to describe precisely energy consumption of WSN node during its operation we designed a specific experimental testbed. The estimation of energy depleted from battery source during node activity is based on the current powering node circuits and especially radio transceiver since it is the main source of energy consumption in the node. We focus mostly on the operations related to communication, which means association of a node to an existing network after node startup and packet transmission. Therefore, we designed an experimental testbed consisting of two communicating nodes IRIS and current measurement setup. The simple WSN topology consists of two nodes: PAN Coordinator and end device. The Coordinator and the end device are identical from the hardware point of view but distinguished by definition of ZigBee roles. As a network nodes, IRIS motes were used. These nodes with XM2110CA module are based on low power Atmel Atmega1281 8bit microcontroller and Atmel AT86RF230 transceiver. Radio part is designed for the radio range 2.4 GHz ISM. These nodes originally come with XMesh protocol stack but we replaced it with Atmel's Bitcloud stack and ported it to IRIS motes for our experiments. The Atmel's Bitcloud stack is full-featured certified ZigBee PRO stack. In addition, the sensor board is equipped with turn-on switch, three LED for user visual communication, serial interface, 10-bit AD converter and other digital interfaces (I2C, SPI). The Coordinator is powered by its battery source as usual, while for powering the end node we used a power supply (3 V) for the experimental purpose. The experimental testbed schema can be seen in Fig. 1.

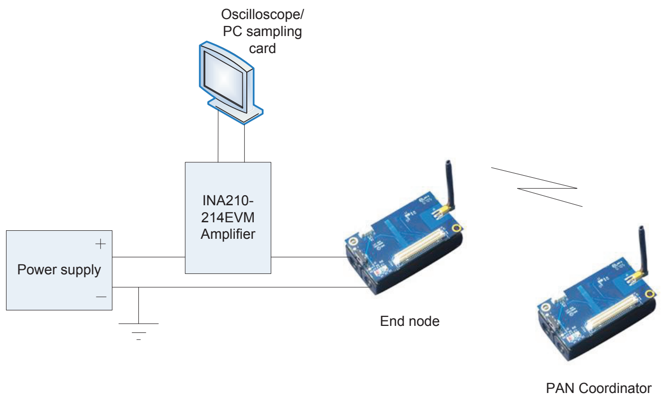

Fig. 1. Experimental testbed for measurement of energy consumption

The main idea of the experiment is to measure a current drained by the end device during communication with the PAN Coordinator. In order to precisely measure the current, a shunt resistor has to be placed in a measurement circuit. The current measurement setup contains a shunt resistor with a known value, which is placed between an energy source and node supply pin. To minimize the impact of the shunt resistor on the node power supply, it has to be very small (we decided for 1 Ω). Because the current passing the shunt resistor is in order of mA the voltage across the shunt resistor has to be amplified for better transparency of results. In our measurement we used INA210 amplifier with the gain 100 especially conceived for current shunt monitoring. See the current shunt monitoring connection in Fig. 2

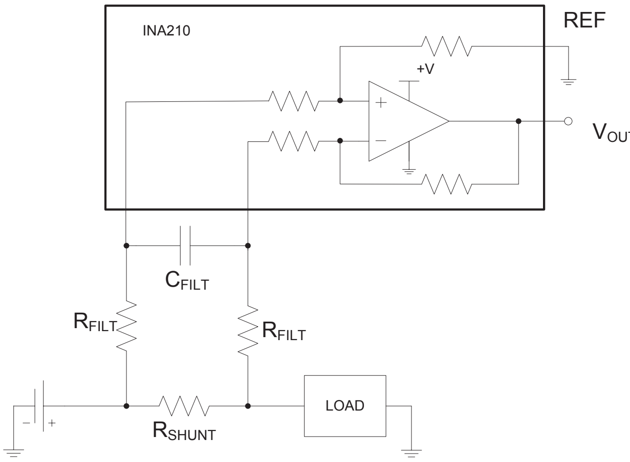

Fig. 2. Current measurement with a shunt resistor and low pass filter(adapted from [14])

There is also a low-pass filter depicted in Fig.1 formed by a capacitor C_{FILT} and a resistor R_{FILT}. The aim of the filter is to cut high frequency changes caused by interference and rapid current state transitions. For the filter design it has to be taken into consideration that a gain of the amplifier can be reduced by high value of R_{FILT} [14].

$$G_{\text{INA210}} = 100 \cdot \frac{R_{\text{INA210}}}{R_{\text{INA210}} + R_{\text{FILT}}} \tag{2}$$

where R_{INA210} is an internal resistance of the amplifier which is 5 kΩ. Another restriction requires that R_{INA210} has to be much higher than R_{SHUNT} in order to avoid negative effect of the filter on the voltage drop. Considering both limitation we choose $R_{\text{FILT}} = 47k\Omega$. According to the chosen value of R_{FILT}, the capacitor C_{FILT} can be calculated based on the cutoff frequency of the filter:

$$f_c = \frac{1}{2\pi \cdot (2R_{FILT}) \cdot C_{FILT}} \tag{3}$$

If we select a value 220 nF for C_{FILT} we get a cutoff frequency 7.7 kHz, which allows us to use a sampling frequency not lower than 15.4 kHz in accordance with the Nyquist theorem.

The last component of the experimental testbed is a measurement and displaying unit. Both fast oscillator and PC sampling card with appropriate software tool can accomplish this function. For our testbed we decided for PC sampling card (Humusoft AD622) and Matlab simulation tool with Simulink and real-time toolbox. The model of measurement in Simulink is in Fig. 3. Block RT Buf In gathers all measured samples which can be stored in buffer for further processing if needed (in case of high load). Values collected during the measurement were stored in a Matlab file and further processed by Matlab in non real-time mode after the experiment.

For the experiment purposes one of the nodes was acting as a PAN Coordinator turned on before the measurement itself in order to create a network, which the end node joins after its startup. The application executed in the end node is a simple function of periodical sending a packet of size 3 Bytes.

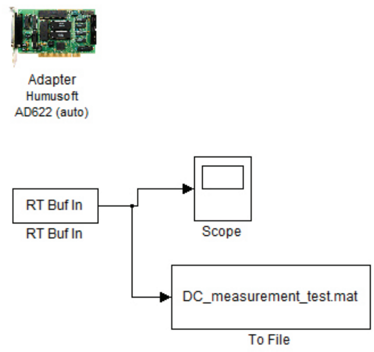

Fig. 3. Using a Matlab real-time library with PCI sampling card

IV. Results and Discussion

In this section we describe the time-line of the experiment and operation of the node after startup in a relation to the current drained from an energy source. Based on the experiment description in the previous part when the end node is turned on its current consumption is as displayed in Fig. 4.

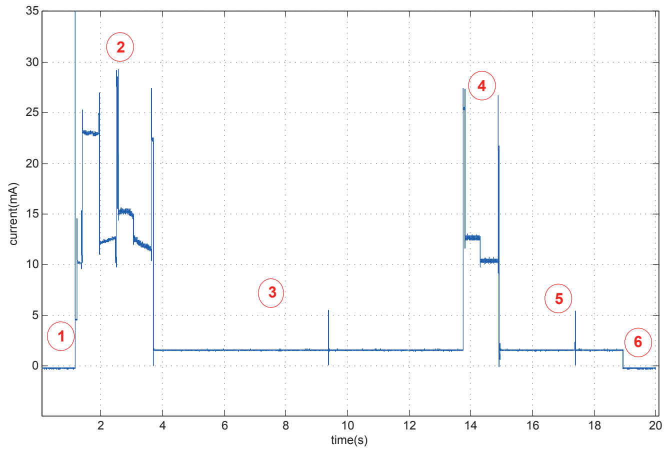

Fig. 4. Radio communication and power consumption of IRIS node

In the phase 1 the node is turned off and the consumption is certainly 0. After turning on the sensor node performs association and binding phase exchanging information with PAN Coordinator and initial packet transmission (phase 2). Then, following the implemented function, the end node waits in a sleep mode (phase 3) draining minimum of energy until periodic packet transmission takes place in phase 4. The required current in a sleep mode is 2 mA approximately, which depends on the sleep mode of microcontroller. In the deepest sleep mode the ATmega1281 requires less than 7.5 μA [13]. Phase 5 represents waiting for next periodic packet transmission in the sleep mode, however, the end node is turned off (phase 6).

The next figure, Fig. 5, represents the turning on process and the association phase only. During the phase 0 the end

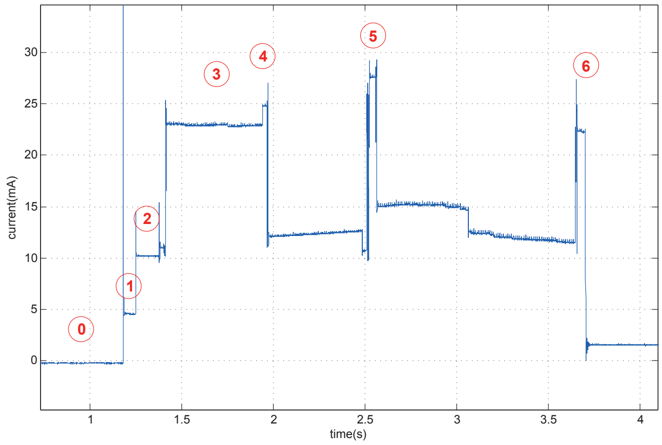

Fig. 5. Association and binding phase of Zigbee communication

node is off. When turned on, the current increases sharply to 50 mA for 0.5 ms. Then, in phase 1, it stays on the level 4.6 mA for approximately 70 ms when system clock is being stabilized. Following phase and level of about 10.2 mA indicates the active state of the microprocessor (phase 2). In the phase 3 the end node sends beacon request and gathers information from PAN Coordinator. Since we already set the communication channel (channel 15) the node then sends rejoin request and gets acknowledgment packet (phase 4). After half a ms several packet transmissions are conducted. The end node gets rejoin response, sends packet with its capabilities as a payload to the Coordinator (the Coordinator spreads them over the network afterwards), sends data request and then the first data packet. Each packet is followed by an acknowledgment at the link layer. It is optional in standard 802.15.4 but ZigBee strictly requires it. It is different with an acknowledgment at the application layer. This ACK is optional in ZigBee communication and user has possibility to change it. In our experiment, the application ACK was required and thus, after approximately one second the end node requests application ACK, which is subsequently received (phase 6).

The application of the end node periodically transmits a data packet after necessary data request command, which is, according to our setting, followed by ACK at the application layer. This sequence is then repeated every ten seconds until the end node is turned off. This means a periodical energy drain, which is imposed by activating the processor and RF circuits transmitting and receiving packets. The time interval of this sequence is 1.18 s. From that period there are two time intervals of 40 ms with the 22 mA current consumption, 0.5 s of 12.5 mA and 0.6 s of 10.3 mA. The current consumption of the activities related to sending of application data is shown in Fig. 6.

As can be seen from all the figures, the energy consumption is mainly dependent on the active state of board circuits. It means mostly processor and RF circuits in our focus. The node association takes about 1.2 s. The half of that time all the circuits consumes energy which means high drain (about 23 mA). When only processor is in the active state, the current is about 12 mA. The important fact is also that while waiting for application ACK the processor is on draining energy (more than 1 s). Therefore, packet transmission and MAC ACK reception itself, when the current increases to 25 mA, is actually miner consumer (takes about 40 ms) in the entire interval of communication.

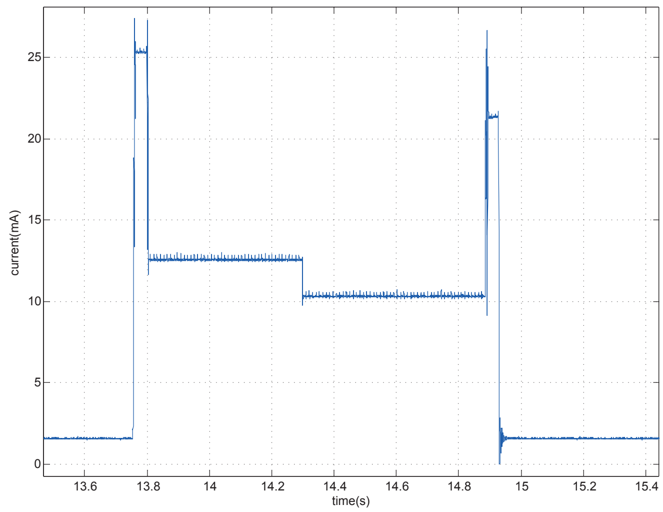

Fig. 6. Data transmission process in ZigBee with acknowledgment at the application layer

V. Conclusion and Future Work

To conclude our work and experiments we focus on the current levels and energy consumption of different phases of communication. First, the start up phase takes approximately 250 ms till the beacon request is sent and the current rises to the value of 23 mA (power supply is 3 V) when RF part is powered. From the node start up the first packet is sent in 1.3 s.

Information how much power is needed for the transmission and reception of a packet is an important input for each energy analysis. The data packet transmission,when maximal transmission power (3 dBm) is used, requires 25 mA (75 mW) and packet reception about 22 mA (66 mW) considering IRIS node. The whole data request and transmission of 3 Bytes of payload takes about 40 ms, about the same takes the request and the reception of application ACK. However, there is a significant interval of waiting. During the time interval of about 1 s, when the processor is on, the current consumption is 13 mA (39 mW) and 10.6 mA (31.8 mW) (see Fig. 6).

Despite the fact that the current drain during the packet transmission and reception is higher, due to the duration of the phases of entire data transmission process (including data request and acknowledgments), more energy is consumed just by processor waiting for the application ACK in active state. This is very important fact, which means that the consumption cannot be calculated only from the transmission rate and number of bytes in the packet but waiting period has to be considered as well. Therefore, we should carefully think about using ACK at the application layer since we have experienced that waiting period takes more than one second in one hop communication and it would be even more for multi hop communication. Moreover, for energy saving, we should set fast MCU oscillator stabilization whenever possible and lower the oscillator frequency if there is no high computational load. General recommendation, which should be obvious, is to make the best of the different sleep modes of microcontroller.

There is several possibilities of energy savings in current WSN ranging from hardware design to energy aware protocols and entire protocol stack. A lot of works evaluating energy consumption is based on the datasheet values or general energy model based on the datasheet information. This work intends to point out that controlling software and implementation of protocols can have considerable effect on the energy consumption and thus, it should be certainly considered as well.

In our future work we will focus on the multi-platform comparison and evaluation of low-power hardware possibilities controlled by firmware. Different manufactures offer WSN platforms with different design which certainly influences the energy consumption. Other broad field of investigation of power consumption is in the firmware controlling the node at the low level and making the best of low-power design strictly respecting its advantages. Especially, taking advantage of various low power modes is one way of prospective energy saving.

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